

Coding & Solution

Date	20 July 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine

Solution Summary

File Overview

File	Purpose
<code>train_model.py</code>	Full ML pipeline — load data, clean, remove outliers (IQR), EDA plots, seasonal analysis, K-Means + Elbow, train & compare Logistic Regression / Random Forest, save best model to <code>model.pkl</code>
<code>app.py</code>	Flask backend — routes for Home (<code>/</code>), About (<code>/about</code>), Find Your Crop (<code>/findyourcrop</code>), and the prediction endpoint (<code>/predict</code>)
<code>templates/home.html</code>	Landing page introducing OptiCrop
<code>templates/about.html</code>	Objectives, methodology, and input features explanation
<code>templates/findyourcrop.html</code>	Input form (N, P, K, temperature, humidity, pH, rainfall) and prediction result display
<code>static/css/style.css, static/js/script.js</code>	Styling and client-side behavior
<code>static/images/*.png</code>	Generated EDA plots (boxplot, correlation heatmap, pairplot, elbow curve)
<code>crop_recommendation.csv</code>	Training dataset (2,200 records, 7 features, 22 crop classes)
<code>model.pkl</code>	Serialized best model + feature order, loaded by <code>app.py</code>
<code>Procfile</code>	Deployment entry point for Gunicorn (<code>web: gunicorn app:app</code>)

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation (module docstrings)	Yes
4	Error handling is implemented for critical operations (invalid/non-numeric input in <code>/predict</code>)	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

`train_model.py` and `app.py` are intentionally decoupled: retraining the model (e.g., after adding more data) only requires re-running `train_model.py` to refresh `model.pkl` — no changes to the Flask app are needed as long as the feature order stays the same.